

Nickolas Whitman

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🌐 <https://nickolasdesigns.github.io/nickolas-whitman-portfolio> • in nickolas-whitman

Work Experience

Associate Automation Software Engineer

Schweitzer Engineering Laboratories Inc.

February 2023 – Present

Pullman, WA

- **Automated Quality Analysis:** Created the X-Ray Inspection Routine which allowed Operators to guarantee sufficient solder volume and positioning of SMT/TH components on a PCB. This technology decreased inspection time by 8,000% and allowed the Production Team to churn through a 2 year inspection backlog in only 3 weeks, saving more than \$2.5 million in operation costs.
- **Driving PCB Testing Processes:** Fathered the ZEUS 2 application which drives the SPEA Flying Probe Tester (FPT) and I/O Conveyor Belt systems. This app also sends Process Control data to SQL and Azure databases, allowing Technicians and Engineers to catch problems with upstream processes before they become an issue. It also hosts a UI for Operators to control the testing system when needed.
- **Automating Project Management:** Developed a Python GUI executable which creates Jira Tasks for various project types, saving 10+ Engineering hours for the Process Engineering Team every week.

Process Engineering Intern

Schweitzer Engineering Laboratories Inc.

January 2022 – January 2023

Pullman, WA

- **Technical Analysis and Client Collaboration:** Conducted thermal profiling and technical analysis for process engineering. Worked with clients in production to design tooling and parts using SOLIDWORKS, rapid prototyping with a 3D Printer, and a streamlined product development life-cycle.
- **Project Management and Development:** Led crucial manufacturing engineering projects, ensuring timely completion and effective team collaboration. Developed Python scripts for data collection, project updates, and management of applications like JIRA, Confluence, and Excel.

SCADA Engineering Intern

Woodhawk Controls

December 2019 – December 2022

Coeur D'Alene, ID

- **Automation and Control Systems:** Followed Electrical Design Blueprints produced by an Electrical Engineer to build various Control Systems designed by Woodhawk Controls and outside customers like Cognex and Amazon.
- **Cognex Project:** Owned the production of Cognex Control Boxes. My responsibilities aimed at ensuring these control systems were built to spec, reporting on weekly production metrics to Company Leadership, and improving efficiency of the production line.

Personal Projects

Automated Trading Bot

Personal Automation Project

January 2019 – Present

- **Bot Performance:** My trading bot produces an average of 3% returns every **month**. A strategy adjustment feature is in active development with the objective of increasing returns based on market sentiment.

Education

B.S. Mechanical Engineering

University of Idaho - Moscow ID

American Society of Mechanical Engineers (ASME), Philanthropy Chair at TKE, Dean's List

B.S. Computer Science

Knox College - Galesburg IL

NCAA DIII Basketball, Engineering Club, Dean's List

Skills & Certifications

Mechanical Engineering.....

Finite Elemental Analysis (FEA): Experience using FEA (ABAQUS/SOLIDWORKS) to validate mechanical specifications are met before creating prototypes.

Certified SOLIDWORKS Associate in Mechanical Design (CSWA): Skilled in creating engineering drawings, assemblies, and 3D printing for rapid prototyping.

American Society of Mechanical Engineers (ASME): Participated in engineering challenges with diverse teams.

Automation/SCADA Engineering.....

Programmable Logic Controllers (PLCs): Ladder Logic, Modbus TCP, Automation Direct PLC, etc.

OSI Model: Experienced with all Layers of the OSI Model (1: Physical, 2: Data Link, 3: Network, 4: Transport, 5: Session, 6: Presentation, 7: Application).

Software Engineering.....

Software Testing: Experience with Black/White/Gray Box, Unit, Integration, System, Acceptance, Functional, Non-Functional, Regression, Smoke, Sanity and Exploratory Testing. I also know how to automate each test level with CI/CD pipelines (Jenkins/Ansible/Github Actions).

Automation: Skilled in automating manufacturing processes to save time and improve efficiency, accuracy and safety of critical manufacturing processes (APIs, Networking, UI Design, back/front end development, etc.).

Data Analysis: Experience with SQL, Azure, PowerBI, Excel VBA Scripts, Splunk.

Programming Languages: C#, Python, C, Java, TypeScript/JavaScript, CSS, HTML, SQL, Go, Bash, PowerShell, and VBA (Excel).

Tech Stacks and Libraries: Dotnet, React, Node JS, Tkinter, ES5, ESNext, WPF, Redis, ASP.NET Core, REST/gRPC/WebSocket APIs, Git, Jenkins, Insomnia, Redis, MongoDB, LinkPad7, Docker, Artifactory.